

2025

(Session : 2024-27)

(Paper ID : 12008)

Time : 3 hours

Full Marks : 80

Candidates are required to give their answers in their own words as far as practicable.

The questions are of equal value.

Answer any five questions.

1. What is linear searching? Write a C program to enter n number in an array. Search a number in the given list of array using liner search.
2. Write a menu driven C program to implement stack operations.
3. What is algorithm complexity? Discuss various methods to assess algorithm performance.
4. What is Bubble Sort? Write an algorithm for bubble sort. What is its complexity? Compare the concept of bubble sort with heap sort and radix sort.
5. What is Queue? Write down the algorithm for enqueue and dequeue operation using link list implementation.
6. What is the role of pointers in dynamic memory allocation? What are the reasons and advantage of allocating memory dynamically.
7. What are Binary Search Trees (BST)? Explain the process of insertion and deletion in BST.
8. What is Double Dimensional array? Write a C program to calculate multiplication of two matrices
9. Explain the concept of double link list? Write a C program function to create, display, count, delete first, delete specific, delete last node.
10. Write short notes on any two of the following:
 - (a) Structure
 - (b) Recursive Function
 - (c) Control structure in C
 - (d) LIFO vs FIFO